

Jie Cao

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INTRODUCTION

My research interests are mainly in millimetre-wave radar systems and artificial intelligence (AI) algorithms in microwave sensing, especially in biomedicine, human-computer interaction (HCI) and indoor localization. For microwave sensing, I am concerned with the construction of machine learning models with good interpretability and high robustness, including the custom pattern recognition models dominated by kernel method and Bayesian inference, and the construction of multimodal big models driven by neural network.

RESEARCH INTERESTS

Microwave sensing, Artificial intelligence algorithm, Microwave biomedical radar system, Contactless cardiopulmonary disease prediction, Human-computer interaction

EDUCATION

Shanghai Jiao Tong University (SJTU)	5 years	Sep. 2023 – Present
Ph.D. in information and communication engineering, advised by Prof. Changzhan Gu		
Harbin Institute of Technology, Shenzhen (HITSZ)	4 years	Aug. 2019 – Jun. 2023
B.E. in communication engineering		

RESEARCH EXPERIENCE

PhD student	Shanghai Jiao Tong University (Shanghai)	
Millimetre-wave radar systems, AI algorithms in microwave sensing		Sep. 2023 – Present
Intern	Shanghai Artificial Intelligence Laboratory (Shanghai)	
Millimeter-wave radar-based non-contact gesture interaction		Aug. 2023 – Jul. 2024
Intern	Open Innovation Lab, DAIKIN (China) Investment Co., Ltd. (Shenzhen)	
Visual recognition algorithms, Biosensor algorithms		Jun. 2023 – Jul. 2023

GRANTS & AWARDS

• May 2026, 2nd Prize of Flash Best Paper Award	IEEE MTT-S IWS, Shenzhen, China
• Apr. 2026, PhD student Initiative Program	IEEE IMBioC, Cosenza, Italy
• Mar. 2026, Ceyear Scholarship	SJTU, Shanghai, China
• Jul. 2025, 2nd Prize of Best Student Paper Award	IEEE MTT-S IMWS-AMP, Wuxi, China
• Jul. 2025, PhD Travel Grant	IEEE MTT-S IMWS-AMP, Wuxi, China
• May 2025, 3rd Prize of Best Student Paper Award	IEEE MTT-S IWS, Xi'an, China
• Sep. 2022, National Scholarship	HITSZ, Shenzhen, China

PROFESSIONAL ACTIVITIES

- Jul. 2026, I became a reviewer for IEEE Transactions on Microwave Theory and Techniques
- Jan. 2026, 2025 Outstanding reviewer for IEEE Transactions on Instrument and Measurement
- Oct. 2025, I became a reviewer for IEEE Journal of Biomedical and Health Informatics
- Aug. 2024, I became a reviewer for IEEE Transactions on Instrument and Measurement

PUBLICATIONS

[1] **J. Cao**, Y. Li, J. Liu, and C. Gu, "A Novel Contactless Concentration Evaluation Technique Using Millimeter-Wave Radar," *2024 International Conference on Microwave and Millimeter Wave Technology (ICMMT)*, Beijing, China, May 16-19, 2024.

[2] **曹杰**, 温力, 唐敏, 郭宜竞, 顾昌展, "基于匹配滤波的非接触生命体征测量," 电子测量技术,

2025,48(01):39-45. (北大核心)

- [3] **J. Cao**, S. Dong, S. Li, X. Wang, M. Zhu and C. Gu, "Microwave Bio-Radar Monitoring of Gene-controlled Activity Changes of Normal and Mouth-breathing Rats," *IEEE MTT-S International Wireless Symposium (IWS)*, Xi'an, China, May 19-22, 2025.
- [4] **J. Cao**, J. Liu, K. Zheng and C. Gu, "Displacement Motion Detection Based on Accurate Phase Accumulation Technique with Reduced Sampling Rate," *IEEE MTT-S International Wireless Symposium (IWS)*, Xi'an, China, May 19-22, 2025. (3rd Prize of Best Student Paper Award)
- [5] W. Gao, L. Meng, **J. Cao**, L. Wen, Z. Zi, Q. Cao, K. Chen and C. Gu, "Noncontact Atrial Fibrillation Detection Utilizing Convolutional Neural Network Based on Doppler Biomedical Radar," *International Conference on Microwave and Millimeter Wave Technology (ICMMT)*, Xi'an, China, May 19-22, 2025.
- [6] L. Wen, **J. Cao**, Z. Zeng, Q. Cao, K. Chen and C. Gu, "Utilizing Biomedical Radar in Clinical Doppler Cardiographic Detection of Atrial Arrhythmias," *IEEE MTT-S International Wireless Symposium (IWS)*, Xi'an, China, May 19-22, 2025.
- [7] **J. Cao**, L. Wen, S. Dong, M. Tang, Y. Guo, C. Shen and C. Gu, "Arrhythmia Classification Based on Distribution Encoding Network with Biomedical Radar," *IEEE MTT-S International Microwave Workshop Series on Advanced Materials and Processes for RF and THz Applications (IMWS-AMP)*, Wuxi, China, Jul. 23-26, 2025. (2nd Prize of Best Student Paper Award)
- [8] **J. Cao**, J. Liu, K. Zheng and C. Gu, "Accurate Phase Accumulation Technique under Vector Cross-multiplication Framework for High-accuracy Displacement Motion Sensing," *IEEE Transactions on Microwave Theory and Techniques*, vol. 73, no. 11, pp. 9279-9291, Nov. 2025. (中科院 1 区)
- [9] Y. Li, G. Xue, **J. Cao**, X. Xu, Q. Chen, L. Zhang and C. Gu, "A Knowledge-Fusion Deep Learning Approach for Premature Beat Classification Using mmWave Radar in Clinical Settings," *IEEE Asia-Pacific Microwave Conference (APMC)*, Jeju Island, Korea, Dec. 2-5, 2025.
- [10] L. Wen, **J. Cao**, Y. Guo, Z. Zeng, Q. Cao, K. Chen and C. Gu, "Robust and Adaptive DC Offset Calibration in Microwave Sensing Systems for Long-Term Microwave Cardiogram Detection," *IEEE Transactions on Microwave Theory and Techniques*, vol. 74, no. 2, pp. 1756-1769, Feb. 2026. (中科院 1 区)
- [11] **J. Cao**, L. Wen, S. Dong, Y. Li, L. Wu, Y. Guo, Z. Zeng, Q. Cao, K. Chen and C. Gu, "Contactless Cardiac Detection of Arrhythmia based on Distribution Encoding Network with Microwave Biomedical Radar," *IEEE Transactions on Microwave Theory and Techniques*, vol. 74, no. 4, pp. 3135-3148, Apr. 2026. (中科院 1 区)
- [12] **J. Cao**, L. Wen, L. Wang and C. Gu, "Contactless Detection of Atrial Fibrillation in Stroke Patients Based on Bayes Probability Model with Microwave Biomedical Radar," *IEEE International Microwave Biomedical Conference (IMBioC)*, Cosenza, Italy, Apr. 26-28, 2026.
- [13] M. Yuan, **J. Cao**, L. Wang and C. Gu "Non-contact Detection of Cheyne-Stokes Respiration and Sighs with Biomedical Microwave Radar," *IEEE MTT-S International Wireless Symposium (IWS)*, Shenzhen, China, May 9-12, 2026. (IEEE MVC Competition Finalist)
- [14] Y. Li, Z. Huang, **J. Cao**, X. Xu, Q. Chen, X. Yuan, J. Chen, L. Zhang, and C. Gu "Non-Contact Cardiac Activity Monitoring via mmWave Radar for Arrhythmia Recognition and STI Estimation," *IEEE MTT-S International Wireless Symposium (IWS)*, Shenzhen, China, May 9-12, 2026. (2nd Prize of Flash Best Paper Award)
- [15] Y. Li, Z. Huang, **J. Cao**, X. Xu, Q. Chen, L. Zhang, and C. Gu "Millimeter-Wave Radar-Based Unobtrusive Arrhythmia Classification Using Deep Learning," *2026 56th European Microwave Conference (EuMC)*, London, UK, Oct. 6-8, 2026.
- [16] Y. Li, **J. Cao**, and C. Gu, "Unobtrusive Cardiac Activity Monitoring Using Millimetre-Wave Radar: Principles, Applications, and Perspectives," *Bioelectromagnetics in Medicine*, 2026.